Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Date:22/8/26

Submitted To:

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Bengaluru – 562106

Experiment No.: 1

Gith hub link: https://github.com/sharvani801/Business-Analytics-Projects

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in Excel files, customer data in CSV files, and product information in another source. Management wants a centralized reporting system. Import data from multiple sources into Power BI, establish the required connections, and prepare the data for business reporting.

Steps Followed

Step 1: Data Collection

- Obtained Sales Data from Excel files.
- Obtained Customer Data from CSV files.
- Obtained Product Data from My SQL Database.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded Excel, CSV files and My SQL data into Power BI.

Step 3: Data Cleaning and Transformation

- Removed duplicate records.
- Handled missing values.
- Renamed columns appropriately.
- Converted data types where necessary.

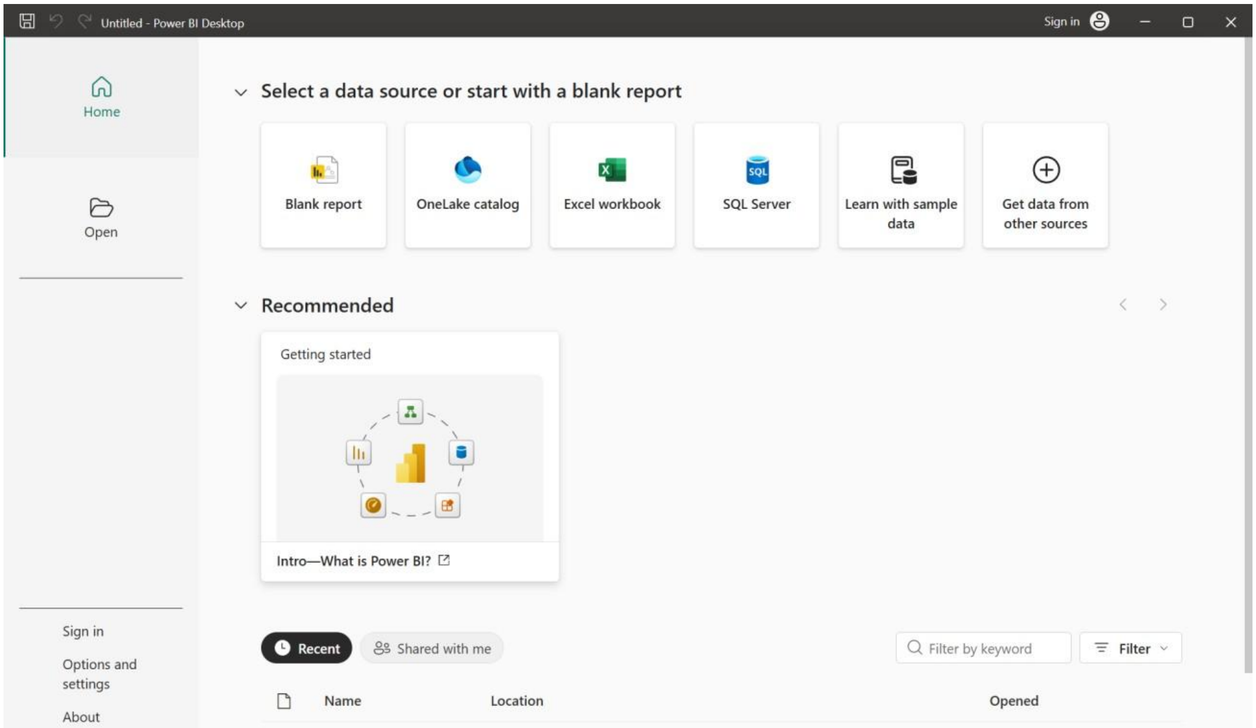
Step 4: Data Modeling

- Created relationships between Sales, Customer, and Product tables.
- Verified one-to-many relationships.

Step 5: Dashboard Development

- Created KPI cards for Total Sales, Orders, and Customers.
- Developed charts for Sales by Category and Region.
- Added slicers for Month, Category, and Region.
- Designed an interactive business dashboard.

Output


Untitled - Power BI Desktop
Sign in
Home
Open
Select a data source or start with a blank report
Blank report
OneLake catalog
Excel workbook
SQL Server
Learn with sample data
Get data from other sources
Recommended
Getting started
Intro—What is Power BI?
Sign in
Options and settings
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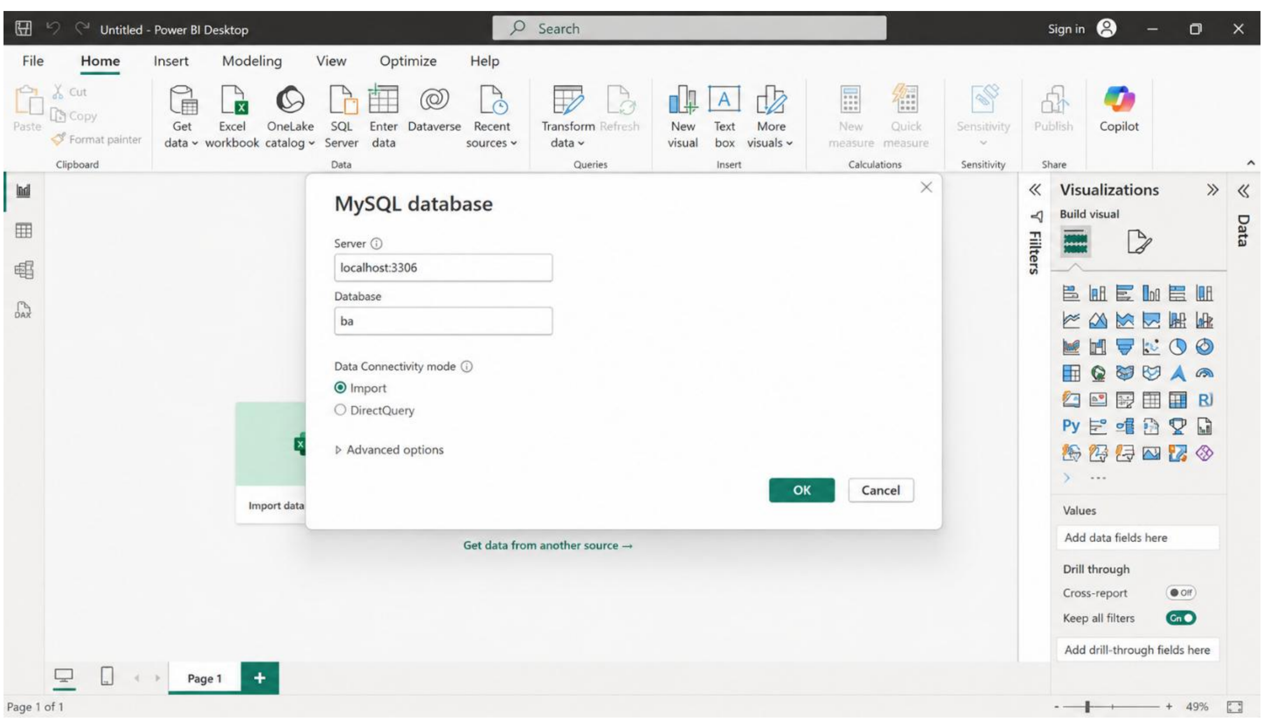
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Page 1 of 1
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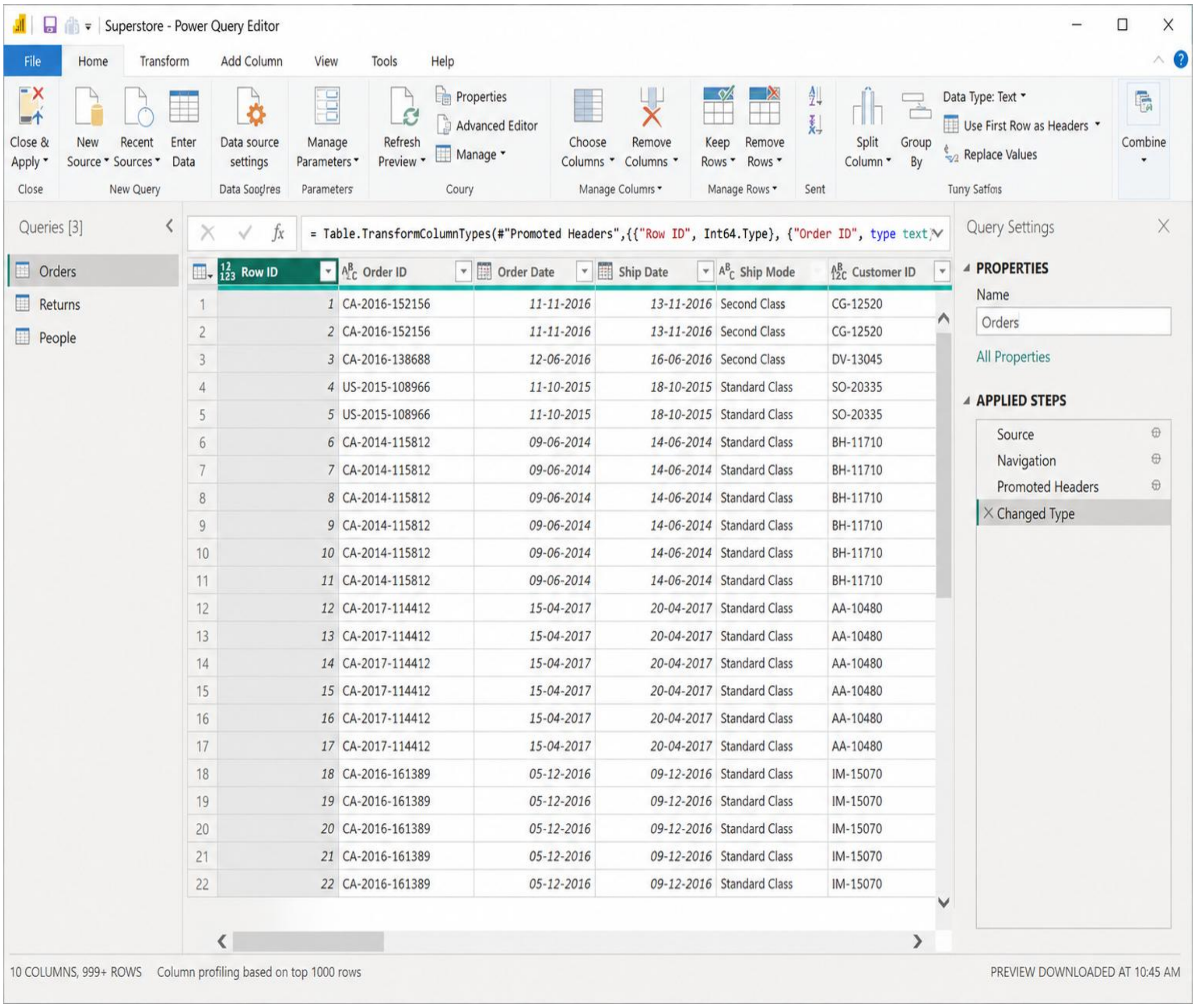
Superstore - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Refresh Preview
Properties
Advanced Editor
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Row ID
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3 CA-2016-138688 12-06-2016 16-06-2016 Second Class DV-13045
4 US-2015-108966 11-10-2015 18-10-2015 Standard Class SO-20335
5 US-2015-108966 11-10-2015 18-10-2015 Standard Class SO-20335
6 CA-2014-115812 09-06-2014 14-06-2014 Standard Class BH-11710
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Query Settings
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Column profiling based on top 1000 rows
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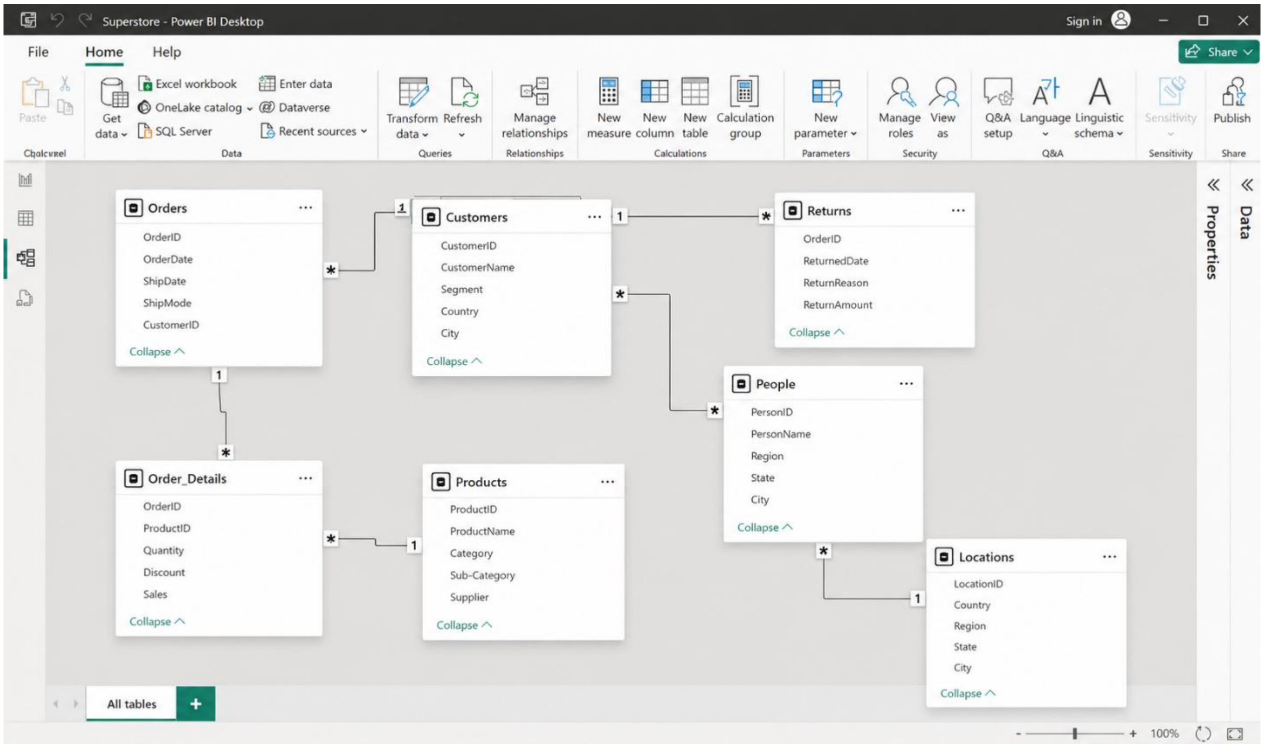
Superstore - Power BI Desktop
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Excel workbook
OneLake catalog
SQL Server
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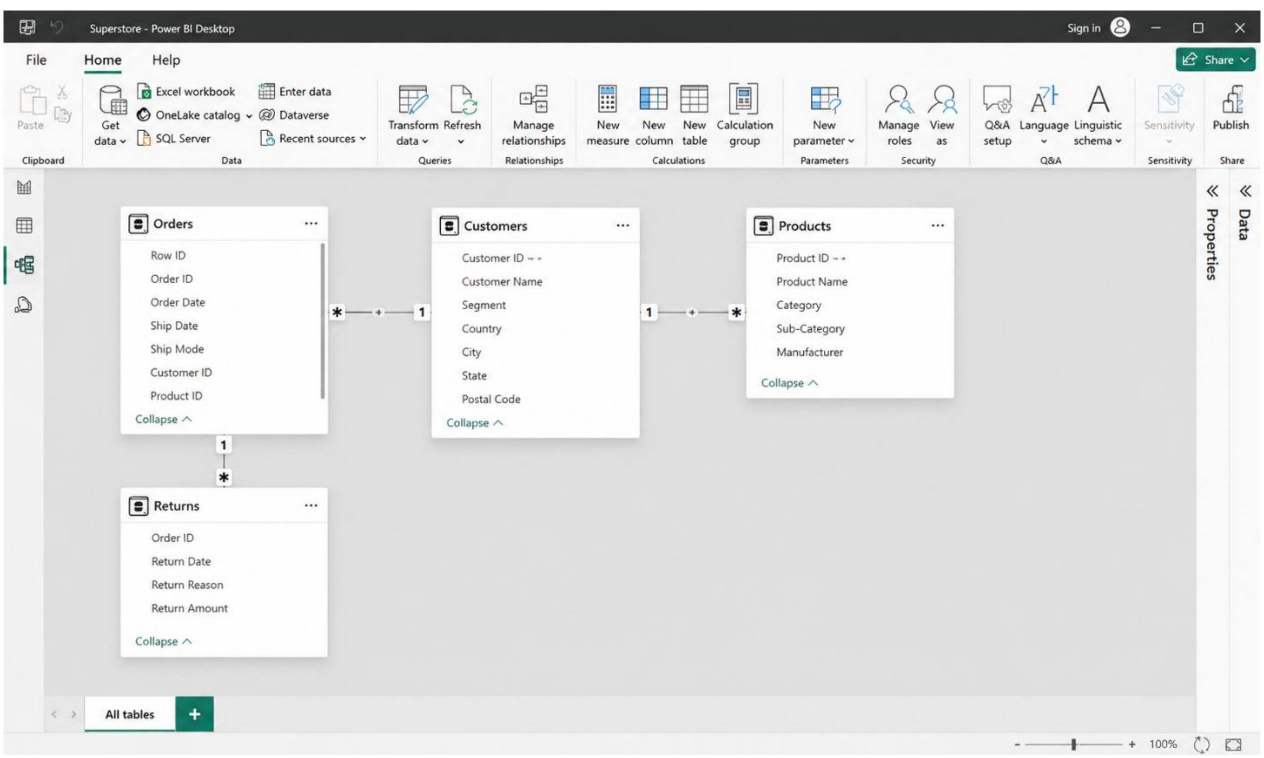
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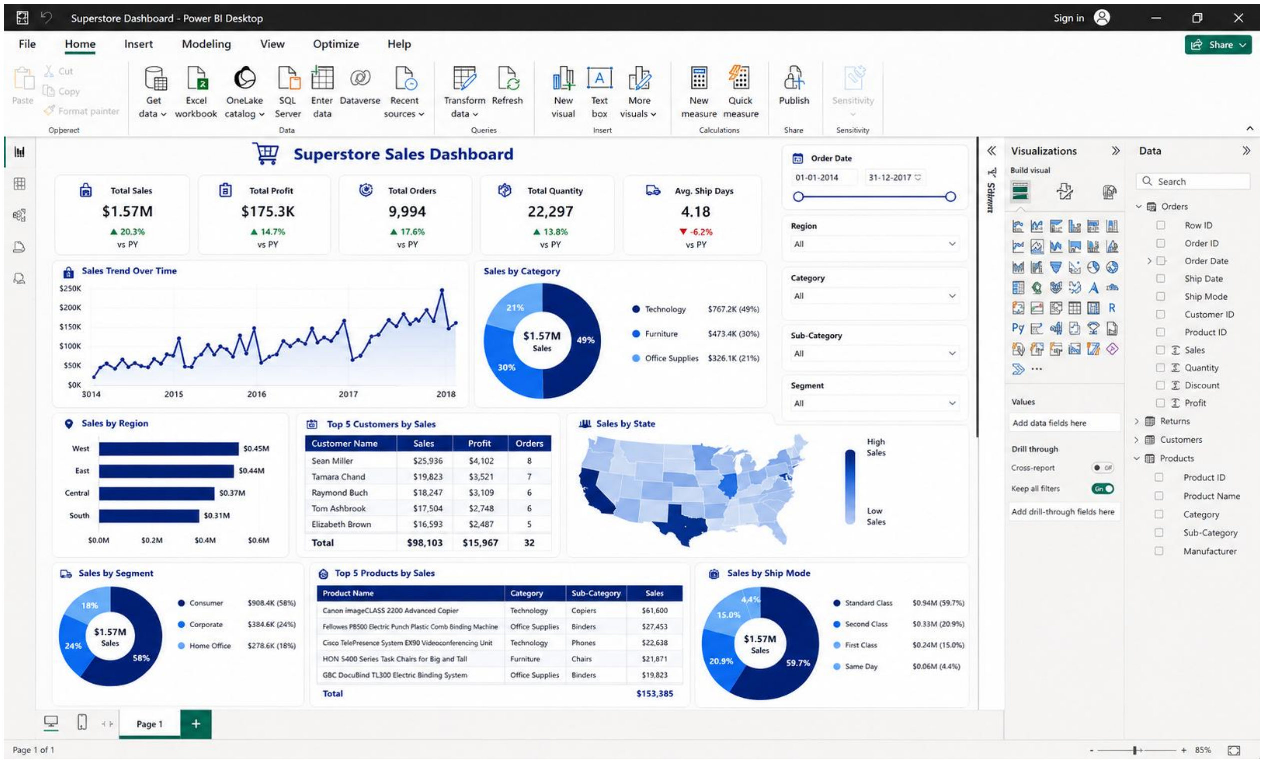

Final Power BI Dashboard – Superstore Dataset

The final Power BI dashboard provides:

- Total Sales and Profit Performance
- Customer-wise Sales Analysis
- Product and Sub-Category Sales Analysis
- Regional and State-wise Sales Performance
- Monthly and Year-wise Sales Trends
- Sales Analysis by Category and Segment
- Order and Quantity Analysis
- Shipping Mode Analysis
- Interactive Filters and Slicers for Region, Category, Sub-Category, Segment, and Order Date

The dashboard enables management to monitor sales, profit, customers, products, regions, and shipping performance and make effective data-driven business decisions using the Superstore dataset.

Conclusion

This project successfully integrated data from multiple sources, including Excel, CSV, and MySQL, into Power BI and transformed the raw data into meaningful business insights. Data cleaning and transformation techniques were applied to improve data quality, including checking data types, handling missing values, and ensuring consistency across the datasets.
The Sales, Customers, and Products tables were connected using appropriate relationships, creating a structured data model for analysis. The developed Power BI dashboard provides a centralized view of business performance through key metrics and visualizations such as total sales, orders, customer information, category-wise performance, regional sales, and monthly sales trends.

Overall, this project provided practical knowledge of data integration, data preparation, data modeling, and business intelligence using Power BI. The resulting dashboard makes the data easier to understand and supports effective analysis and better data-driven decision-making.

Learning Outcomes

Through this project, I learned:

- Understand how to import and integrate data from multiple sources, including Excel, CSV, and MySQL, into Power BI.
- Learn how to clean and transform raw data by changing data types, handling missing values, and checking for duplicate records.
- Understand the importance of data modeling and create appropriate relationships between Sales, Customers, and Products tables.
- Gain practical knowledge of creating DAX measures such as Total Sales, Total Orders, and Total Customers.
- Learn how to create and use interactive visualizations, charts, KPIs, and slicers in Power BI.
- Develop the ability to analyze sales performance, customer information, product categories, regional performance, and monthly trends.
- Understand how business data can be transformed into meaningful insights to support data-driven decision-making.
- Gain practical experience in using Power BI as a business intelligence and data visualization tool.